

Problem Statement



AI Agent for Academic Guidance: This project aims to develop an AI agent for the IITM BS SEEK portal, enhancing student engagement through course-specific guidance while ensuring academic integrity. Using Retrieval-Augmented Generation (RAG), the agent will offer study strategies and direct students to resources without providing direct answers. The development includes six milestones, covering user needs, UI design, system architecture, API development, testing, and final integration. The end goal is a working prototype seamlessly integrated with SEEK.

User Categories & Roles

Primary Users: Students

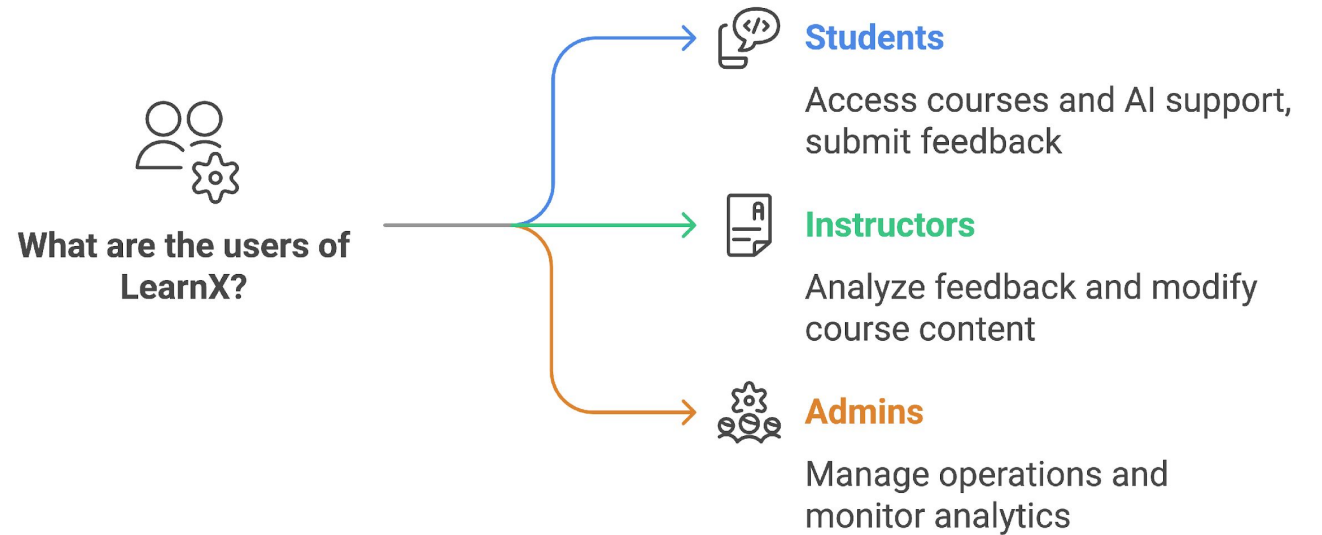
- Access courses, assignments, and AI chatbot support
- Submit course feedback

Secondary Users: Instructors

- View and analyze student feedback
- Create or Modify course contents

Secondary Users: Admins

- Manage platform operations (course creation, user roles)
- Monitor feedback analytics



User Stories

As a Student, I should be able to:

- **Summarize Lectures:** Quickly access concise summaries of each session, highlighting key points for efficient revision.
- **Access a Question Bank:** Browse a dynamic question bank filled with practice questions modeled after previous term's papers to aid in comprehensive exam preparation.
- **Ask Course-Related Questions:** Directly interact with the AI-driven chatbot for instant answers to queries related to course materials, grading policies, and more.
- **Receive Coding Assistance:** Get support with coding tasks and practice coding solutions independently, enhancing my problem-solving skills.
- **Get Assignment Reminders:** Receive automated notifications about upcoming assignments and deadlines, ensuring timely submissions and better time management.

As a Course Instructor, I should be able to:

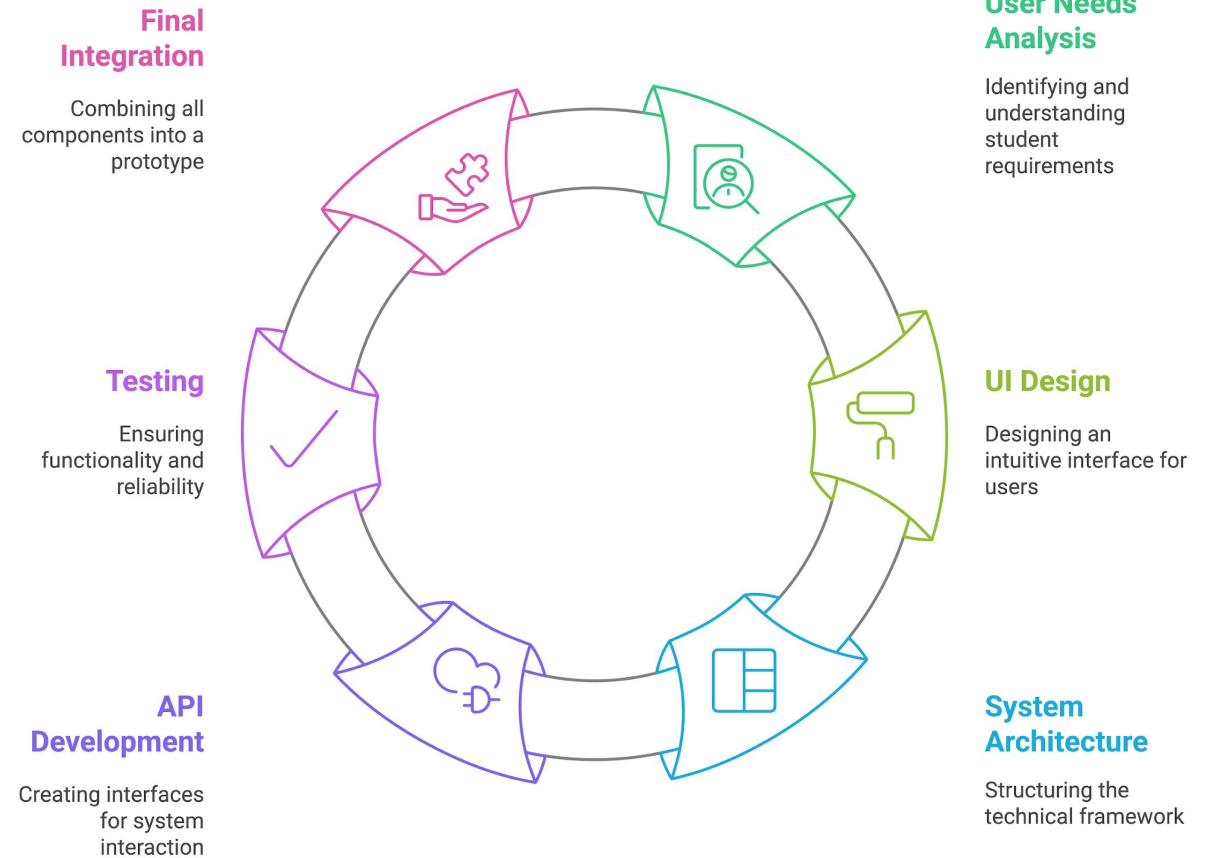
- **Upload Course Materials:** Seamlessly upload weekly course content and reference materials, making it accessible to students.
- **Create Assignments and Activities:** Design engaging weekly assignments, activities, and practice questions to enhance learning.
- **Review Student Feedback:** Analyze student feedback using sentiment analysis and keyword extraction to identify areas for improvement and implement actionable insights within two weeks.
- **Send Updates:** Keep students informed through timely course schedule updates, content changes, and announcements via email or the portal.

As a Portal Admin, I should be able to:

- **Assign Instructors:** Manage instructor access and roles within the platform to ensure proper content management.
- **Review Course Feedback:** Analyze course feedback through sentiment analysis, keyword extraction, and root cause analysis to pinpoint issues and improve course delivery.
- **Manage Course Data:** Make necessary updates to admin-related course data to ensure smooth operations across the platform.

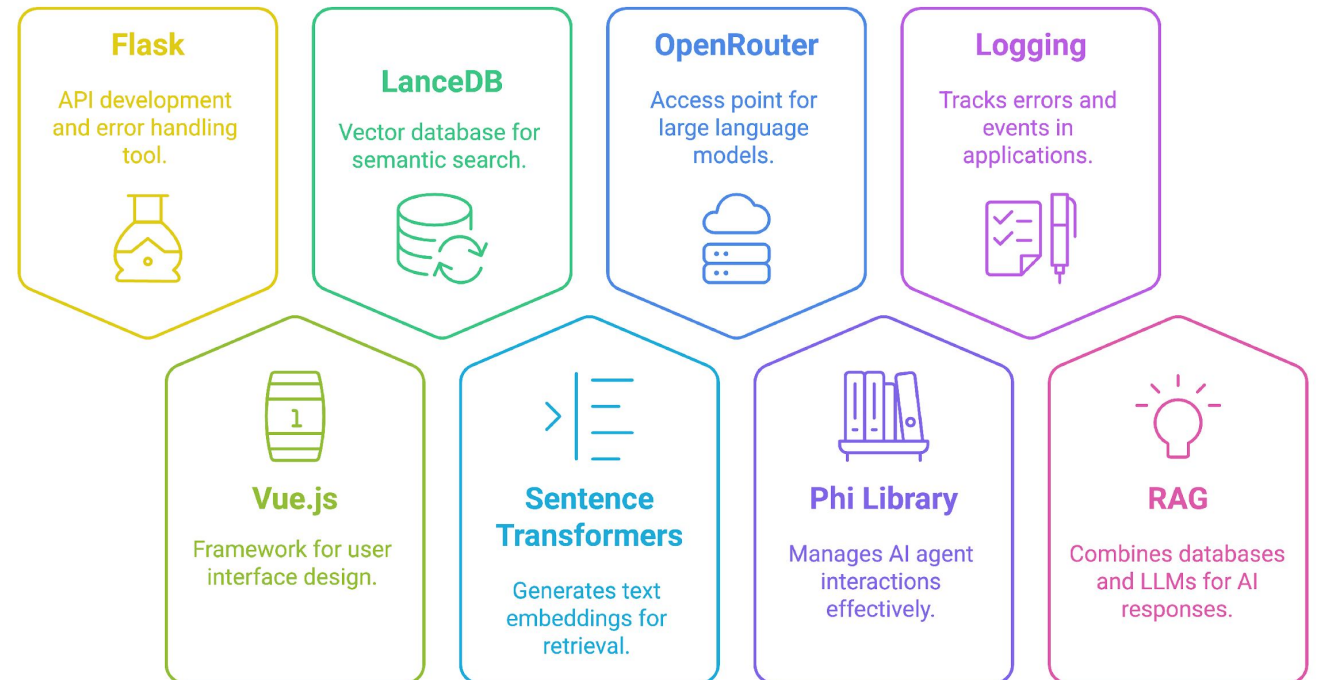
Our Development Strategy

AI Agent Development Overview



Tech-Stack

Technology Components



Project Management

- Jira - A cloud-based project management tool enabling teams to track tasks and bugs, with role-based permissions and automation for secure, efficient collaboration

Projects / SE_Project'25 - Team22
Milestone 3 Sprint 2

The screenshot displays a Jira project dashboard for 'SE_Project'25 - Team22, specifically for 'Milestone 3 Sprint 2'. The interface includes a search bar, a list of team members (IB, PN, DR, +2), and a dropdown menu for 'Epic'. The main area is a Kanban board with three columns: 'TO DO', 'IN PROGRESS 5', and 'DONE 9'. The 'TO DO' column has a '+ Create issue' button. The 'IN PROGRESS' column contains five tasks: 'Chat with Video Feature' (GEN AI, SCRUM-17, 2, IB), 'Dashboard UI Modifications' (FRONTEND, SCRUM-26, 2, PN), 'CronJob Deadline Reminders' (CRONJOB, SCRUM-19, 1, SB), 'YAML Documentation & Validation of all the APIs' (SCRUM-44, 1, IB), and 'Gen AI Prototype - Chatbot for Courses' (GEN AI). The 'DONE' column contains four tasks: 'Dashboard Modifications' (BACKEND, SCRUM-22, 2, IB), 'Storyboard Creation for Milestone 2' (SCRUM-23, 1, IB), 'Course Management Dashboard' (FRONTEND, SCRUM-39, 1, IB), and 'Frontend for Assignments' (FRONTEND, SCRUM-38, 1, PN). A 'Quickstart' button is visible in the bottom right corner.

Projects / SE_Project'25 - Team22
Milestone 3 Sprint 2

Search

IB PN DR +2 Epic

GROUP BY None Insights View settings

TO DO

+ Create issue

IN PROGRESS 5

Chat with Video Feature
GEN AI
SCRUM-17 2 IB

Dashboard UI Modifications
FRONTEND
SCRUM-26 2 PN

CronJob Deadline Reminders
CRONJOB
SCRUM-19 1 SB

YAML Documentation & Validation of all the APIs
SCRUM-44 1 IB

Gen AI Prototype - Chatbot for Courses
GEN AI

DONE 9

Dashboard Modifications
BACKEND
SCRUM-22 2 IB

Storyboard Creation for Milestone 2
SCRUM-23 1 IB

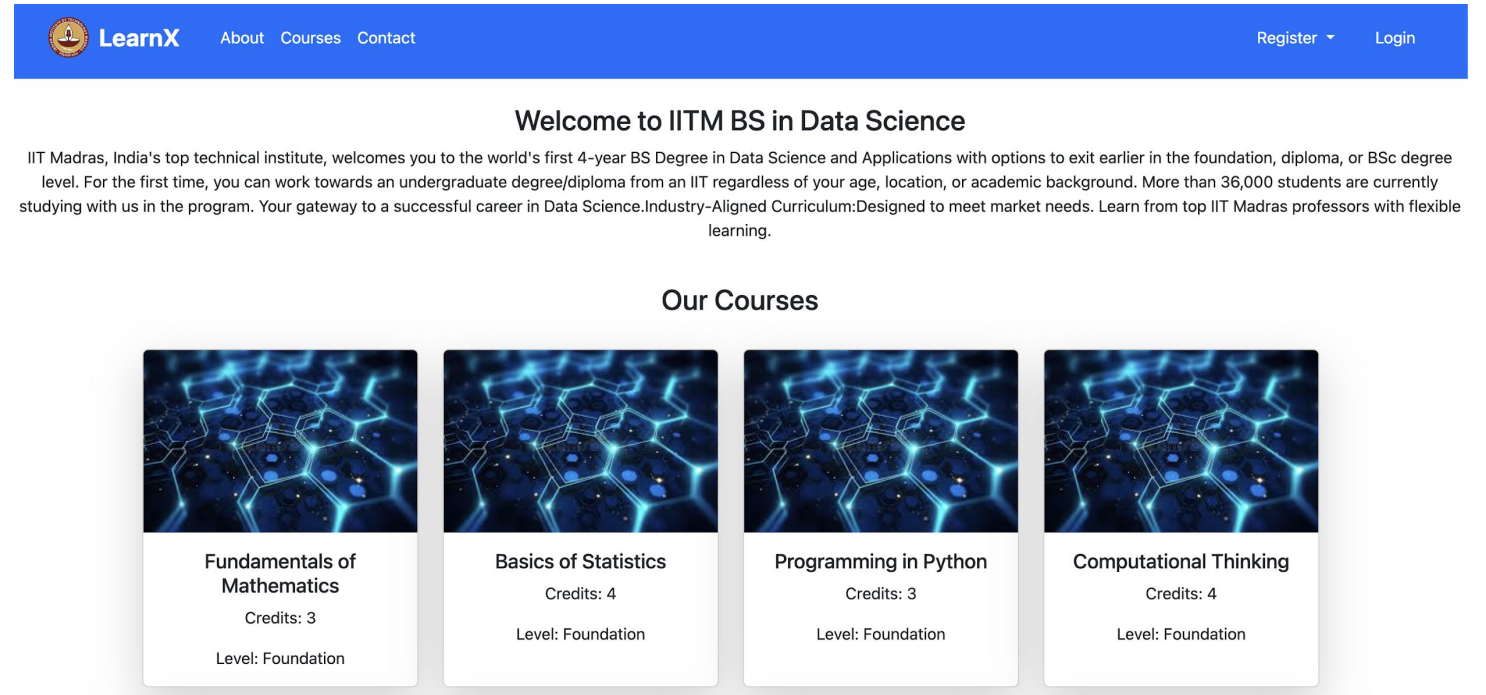
Course Management Dashboard
FRONTEND
SCRUM-39 1 IB

Frontend for Assignments
FRONTEND
SCRUM-38 1 PN

Quickstart

Landing Page

The entry point for users, featuring intuitive navigation, a search bar, and AI-powered assistance for seamless interaction.



Student Dashboard

A personalized hub showcasing enrolled courses with an interactive feedback feature to enhance learning experiences.



LearnX

Dashboard

Welcome, Dummy Student

[Logout](#)



Advanced Data Structures & Algorithms

Credits: 5

Term: May 2025



MLF

Credits: 2

Term: May 2025

Give Feedback

Course Pages – Student View

Access weekly course materials, assignments, and programming tasks, with an AI-powered chatbot for instant content-based support.

Week 1

Lecture 1.1
Lecture 1.2
Lecture 1.3
Lecture 1.4

Assignments:
Graded Assignment 1.1

Week 2

Programming Assignment

Course Content

Select a week from the sidebar to view the lectures and assignments.

What is Machine Learning?

Watch later

Share

1/94

What is Machine Learning?

Prof Harish Guruprasad Ramaswamy

Assistant Professor, Department of Computer Sciences & Engineering,
IIT Madras

Watch on

YouTube

Help

Ask for Help

Week 1

Lecture 1.1
Lecture 1.2
Lecture 1.3
Lecture 1.4

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Assistant Professor, Department of Computer Sciences & Engineering,
IIT Madras

Watch on YouTube

Ask about this course...

Send

Previous Interactions:

Q: What is SVM? How does it work?

A: ## Summary

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification, regression, and outliers detection. The key idea of SVM is to find the optimal boundary (hyperplane) that best separates the data into different classes.

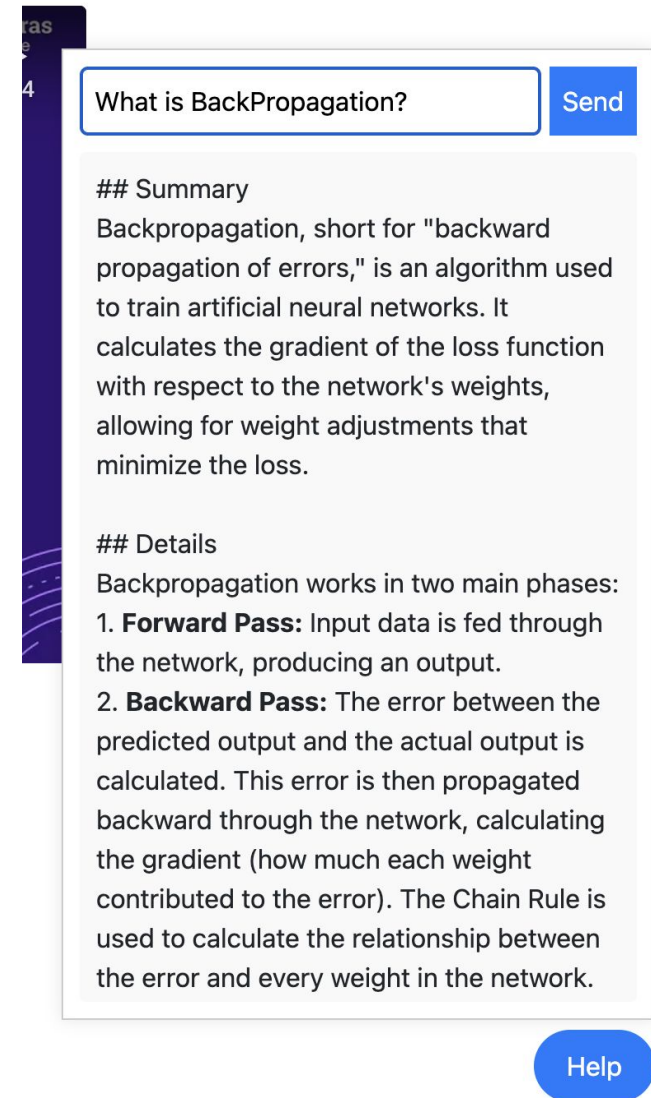
Details

SVM in Classification: SVM is primarily used for classification tasks. Given labeled training data, the algorithm finds the hyperplane that maximizes the margin between the classes. The margin is the distance between the hyperplane and the closest data points from either class, known as support vectors.

Help

The ChatBot - TutorX

The chatbot provides course-related answers, resolves student doubts, and offers instant, AI-driven academic support.



What is BackPropagation? Send

Summary

Backpropagation, short for "backward propagation of errors," is an algorithm used to train artificial neural networks. It calculates the gradient of the loss function with respect to the network's weights, allowing for weight adjustments that minimize the loss.

Details

Backpropagation works in two main phases:

1. **Forward Pass:** Input data is fed through the network, producing an output.
2. **Backward Pass:** The error between the predicted output and the actual output is calculated. This error is then propagated backward through the network, calculating the gradient (how much each weight contributed to the error). The Chain Rule is used to calculate the relationship between the error and every weight in the network.

Help

AI Chatbot with Context Awareness - TutorX

Our chatbot retains past conversations, maintaining context across multiple interactions to provide more relevant and personalized responses.

What is Exploding Gradient Problem

Send

Previous Interactions:

Q: Explain backprop

A: ## Summary

Backpropagation, short for "backward propagation of errors," is an algorithm used to train artificial neural networks. It calculates the gradient of the loss function with respect to the network's weights, allowing for weight adjustments that minimize the loss.

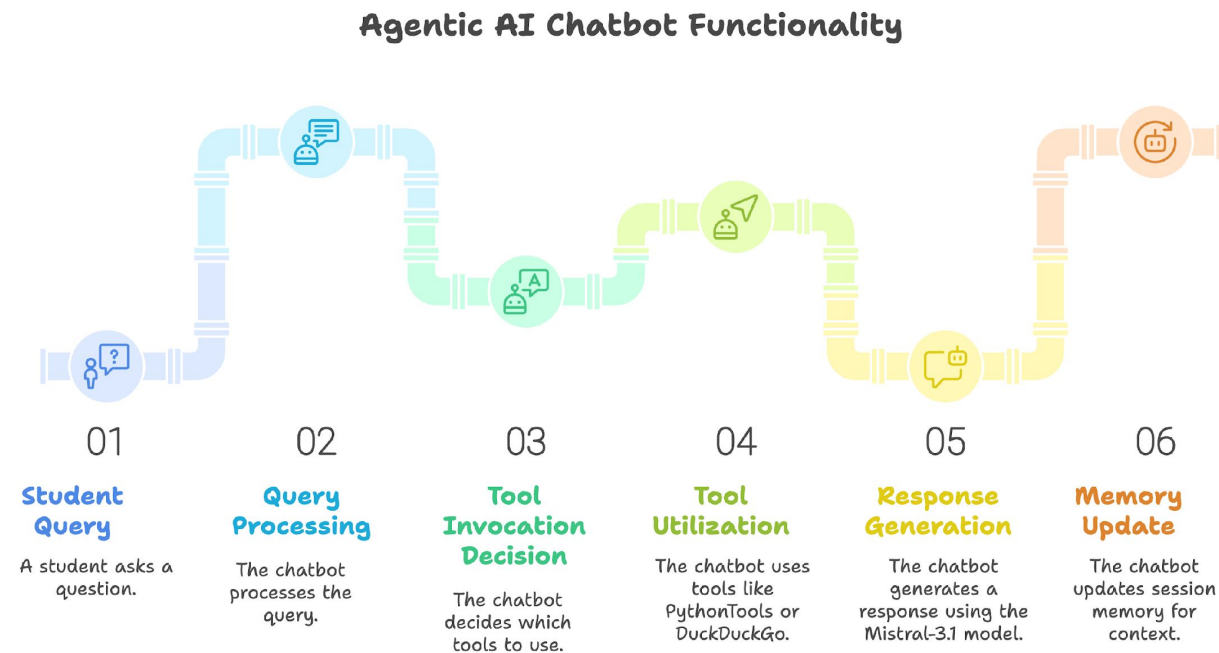
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Help

Deep Dive into LearnX: The Agentic AI Chatbot



Salient Features:

- **Agentic Intelligence:** Autonomously interacts and decides when to use tools like **PythonTools** and **DuckDuckGo** for enhanced, context-aware responses.
- **Course-Specific Support:** Uses **LanceDB** and **Sentence Transformers** to retrieve and generate relevant course-related answers.
- **Mistral-3.1 (24B Instruct):** Powered by a state-of-the-art LLM for accurate, academically-driven responses.
- **Memory & Persistence:** Stores and recalls previous interactions for continuous context, providing seamless, personalized support.
- **Tool Integration:** Leverages a hybrid system combining **vector search** and external tools for detailed problem-solving and learning assistance.



Advanced Data Structures & Algorithms

Credits: 5

Term: May 2025



ML

Cre

Ter

Submit Feedback



Select Course:

Advanced Data Structures & Algorithms



Your Feedback:

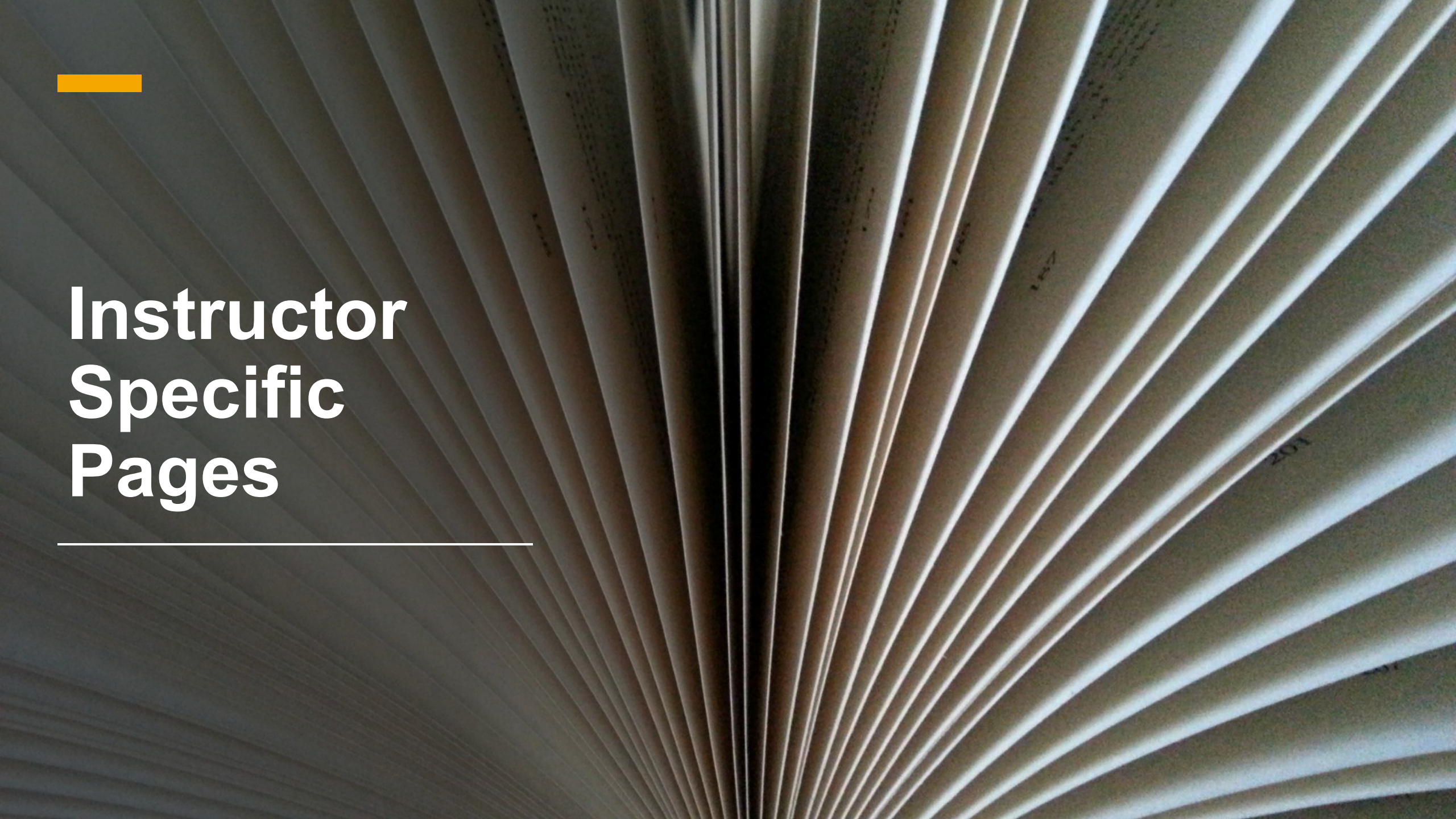
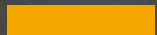
Write your feedback here

Close

Submit

Feedback Mechanism

Students can share feedback on courses, assignments, and overall learning experience, helping improve content and teaching strategies.



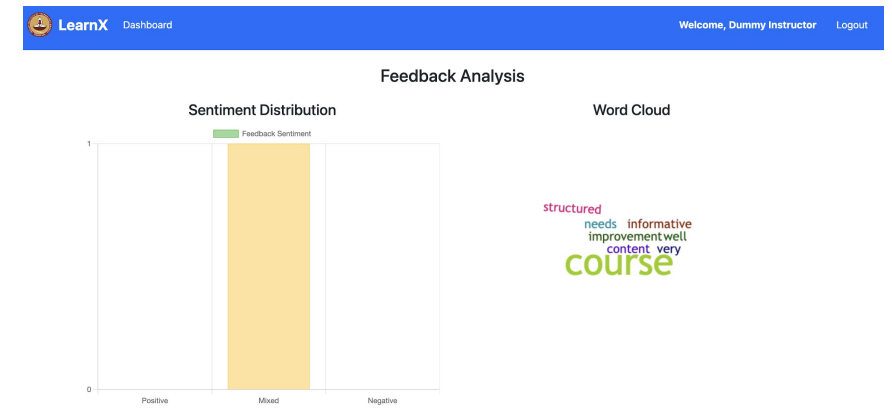
Instructor Specific Pages

Instructor Feedback Dashboard – AI-Powered Insights

This page provides instructors with an AI-driven analysis of student feedback. It features:

- **Sentiment Distribution:** A breakdown of feedback into positive, mixed, and negative categories.
- **Word Cloud:** A GenAI-generated visualization highlighting key themes and recurring topics from student feedback.

This helps instructors quickly gauge student sentiments and identify areas for course improvement.



Course Management

Course Content Management

Week 1

Manage Lectures & Assignments for Week 1

Week 2

Manage Lectures & Assignments for Week 2

Week 3

Manage Lectures & Assignments for Week 3

Lectures

Add Another Lecture

Assignments

Add Assignment

Add Another Week Submit

Key Features:

- **Week-Based Organization:** Instructors can manage lectures and assignments for each week separately.
- **Lecture & Assignment Management:**
 - Add new lectures and assignments dynamically.
- **Scalability:**
 - Option to add more weeks as the course progresses.
- **User-Friendly Design:**
 - Clear buttons for quick content updates and submissions.



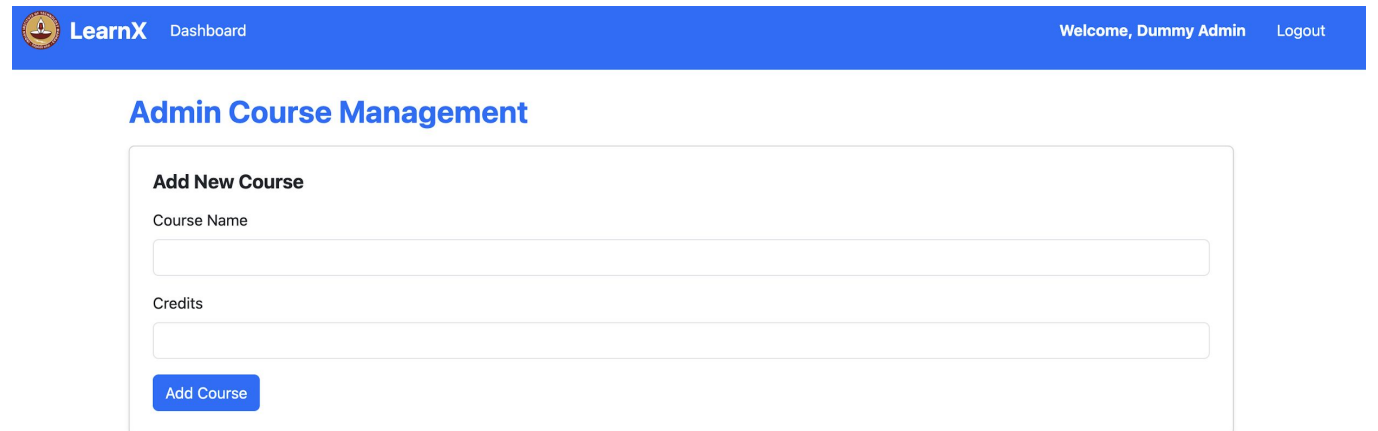
Admin Specific Pages

Admin Course Management

A streamlined interface for administrators to manage courses in **LearnX**.

- **Key Features:**

- **Add New Courses** with name and credit details.
- **User-Friendly Interface** for quick course creation.
- **Secure Admin Access** with authentication and session management.



The screenshot displays the 'Admin Course Management' page within the 'LearnX' dashboard. The top navigation bar is blue, featuring the 'LearnX' logo, a 'Dashboard' link, and a user greeting 'Welcome, Dummy Admin' with a 'Logout' link. The main content area has a heading 'Admin Course Management' in blue. Below this is a form titled 'Add New Course'. The form contains two input fields: 'Course Name' and 'Credits'. At the bottom of the form is a blue button labeled 'Add Course'.

Course Management - Edit

Courses List



Advanced Data Structures & Algorithms

Credits: 5
Instructor ID: 2

Edit

Edit Course

Course Name

Advanced Data Structures & Algorithms

Credits

5

Instructor ID

2

Cancel

Save Changes



SE

Credits: 5
Instructor ID: 6

Edit



SC

Credits: 5
Instructor ID: 7

Edit



ST

Credits: 5
Instructor ID: Not Assigned

Edit



DLP

Credits: 5
Instructor ID: Not Assigned

Edit



DL

Credits: 5
Instructor ID: Not Assigned

Edit

Effortlessly update course details in **LearnX** with an intuitive edit interface.

•Key Functionalities:

- **Modify Course Name, Credits, and Instructor ID** dynamically.
- **Instant Updates** with a **Save Changes** button.
- **Seamless Admin Control** for managing course assignments.

Course Sentiment Analysis Powered by GenAI

This sentiment analysis is generated using **cutting edge Generative AI (GenAI) Large Language Models (LLMs)**. Specifically, we utilized **the latest Gemma model (Gemma 2B - March 2024 release)** to analyze course feedback and determine sentiment.

AI-Powered Insights:

- "Advanced Data Structures & Algorithms" → **Mixed** Sentiment
- "MLF" → **Positive** Sentiment

Course Sentiment Analysis

Advanced Data Structures & Algorithms

Sentiment: **Mixed**

MLF

Sentiment: **Positive**

CronJobs

